AI.304

Robot Operating Syste

COURSE OVERVIEW

The ROS ecosystem: nodes, topics, services, actions, launch files, and middleware for robot software.



ILLUSTRATIVE HERO IMAGERY

ROBOT OPERATING SYSTEMS

1

SESSION

 60 MINUTES

ORIENTATION: THE ROBOT OS

KEY TOPICS

- What is ROS
- ROS architecture
- Why ROS dominates robotics
- ROS1 vs ROS2

TEACHING FOCUS

Learn the middleware of modern robots.

▲ **KEY TAKEAWAY** ROS is the shared language of robot software.

ROBOT OPERATING SYSTEMS

2

SESSION

⏱ 60 MINUTES

NODES & COMMUNICATION

KEY TOPICS

- ROS nodes
- Topics and publishers/subscribers
- Services
- Messages and parameters

TEACHING FOCUS

Build modular robot software.

▲ **KEY TAKEAWAY** Nodes talk through topics and services.

ROBOT OPERATING SYSTEMS

3

SESSION

60 MINUTES

TOOLS & WORKFLOW

KEY TOPICS

- roscore, rosrun, roslaunch
- rviz and gazebo
- rosbag
- Catkin/colcon build

TEACHING FOCUS

Develop, visualize, and log robot software.

KEY TAKEAWAY Tools make ROS development tractable.

4

SESSION

60 MINUTES

ACTIONS & COORDINATION

KEY TOPICS

- Actionlib and action servers
- Goal feedback and preemption
- Complex robot behaviors
- State machines (SMACH)

TEACHING FOCUS

Manage long-running robot tasks.

KEY TAKEAWAY Actions coordinate multi-step behavior.

5

SESSION

60 MINUTES

ROS IN PRACTICE

KEY TOPICS

- Integrating drivers
- Multi-robot namespaces
- Real-time considerations
- Debugging ROS

TEACHING FOCUS

Deploy ROS on real robots.

KEY TAKEAWAY ROS must run reliably on hardware.

ROBOT OPERATING SYSTEMS

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SESSION

⏱ 60 MINUTES

ROS PROJECT & REVIEW

KEY TOPICS

- ROS-based robot demo
- Course recap and assessment
- Toward cognitive robotics (AI.305)
- Code review

TEACHING FOCUS

Build a complete ROS robot system.

▲ **KEY TAKEAWAY** ROS integrates perception, planning, and action.